

CIUDAD DEL ESTE GUARANI, Paraguay

WMO: 862183

Lat: 25.450S

Lon: 54.833W

Elev: 866

StdP: 14.24

Time Zone: -4.00 (PRY)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.0	44.5	35.2	31.0	48.6	37.8	34.4	49.6	23.3	70.3	21.1	72.9	4.9	180	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.2	97.0	74.8	95.1	74.8	93.2	75.0	80.0	89.8	79.0	88.7	78.1	87.6	10.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	146.8	85.3	76.6	143.1	84.4	75.3	137.2	82.7	44.5	89.7	43.2	89.3	42.1	88.5	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.2	18.6	16.6	DB	35.0	101.5	2.6	1.4	33.2	102.5	31.7	103.3	30.2	104.1	28.4	105.1	
			WB	33.3	83.0	2.6	1.6	31.4	84.2	29.8	85.1	28.3	86.0	26.4	87.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.9	80.5	80.1	78.5	73.9	66.4	64.4	63.6	67.1	70.1	74.9	76.4	78.9	(6)
	DBStd	8.68	3.26	3.55	4.23	6.14	6.48	7.61	8.20	8.78	8.34	5.80	4.65	4.04	(7)
	HDD50	13	0	0	0	0	1	3	6	3	0	0	0	0	(8)
	HDD65	437	0	0	0	11	62	104	128	87	43	2	0	0	(9)
	CDD50	8356	946	843	884	717	508	435	427	533	604	772	791	896	(10)
	CDD65	3305	481	423	419	278	104	86	84	152	196	309	342	431	(11)
	CDH74	29751	4749	3986	3793	2153	606	477	587	1341	1898	2799	3237	4125	(12)
	CDH80	12839	2186	1829	1692	825	150	94	141	526	873	1265	1421	1837	(13)
(14) Wind	WSAvg	7.4	6.9	6.8	6.8	6.7	6.7	7.0	7.5	8.5	8.7	8.4	7.8	7.3	(14)
(15) Precipitation	PrecAvg	68.6	7.1	5.8	5.0	5.9	6.4	4.2	3.9	4.0	5.4	7.0	7.5	7.0	(15)
	PrecMax	97.2	15.1	15.8	13.7	11.7	19.3	12.5	11.9	13.6	12.6	11.1	14.6	18.1	(16)
	PrecMin	47.4	2.5	0.0	1.0	2.4	0.0	0.0	0.5	1.2	0.7	0.4	2.3	0.0	(17)
	PrecStd	14.7	2.8	3.8	3.6	2.2	4.7	3.1	2.7	2.9	3.3	3.0	3.6	4.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.3	99.0	97.1	94.0	88.0	85.8	86.4	93.1	98.6	100.1	97.1	97.1	(19)
		MCWB	76.5	76.1	75.0	76.0	72.0	69.4	68.2	69.5	72.0	76.0	74.7	75.6	(20)
	2%	DB	94.9	95.4	94.6	91.2	84.3	82.6	84.1	89.5	94.6	95.1	93.6	94.7	(21)
		MCWB	76.4	76.3	75.1	74.7	71.4	69.1	67.7	68.4	69.8	74.3	74.0	75.7	(22)
	5%	DB	92.8	92.9	91.6	88.1	80.7	80.2	80.7	86.0	89.7	91.3	91.2	91.7	(23)
		MCWB	76.4	76.2	75.0	73.7	69.6	68.3	67.4	67.7	69.7	72.8	73.6	75.8	(24)
	10%	DB	90.0	89.8	89.3	85.6	77.1	75.9	77.0	82.2	84.5	87.4	88.0	89.5	(25)
		MCWB	76.2	75.2	74.4	73.0	67.9	67.4	66.0	67.0	67.8	71.5	72.6	75.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.6	81.1	79.8	79.5	75.8	72.9	72.0	73.5	75.6	79.2	80.4	80.9	(27)
		MCDB	90.5	91.3	89.5	88.8	83.2	78.8	80.9	85.8	91.0	93.6	90.3	90.5	(28)
	2%	WB	79.9	79.2	78.1	76.9	73.2	70.8	69.9	71.0	73.3	76.6	77.9	79.3	(29)
		MCDB	89.4	89.1	87.9	85.8	80.1	78.2	79.7	82.7	87.3	89.5	88.1	88.5	(30)
	5%	WB	78.7	78.1	77.0	75.5	71.4	69.3	68.1	69.3	71.2	75.0	76.1	78.1	(31)
		MCDB	88.1	87.7	86.7	83.9	77.4	77.0	77.2	81.6	82.6	86.8	86.1	87.0	(32)
	10%	WB	77.6	77.0	75.9	74.0	69.7	67.7	66.7	67.7	69.7	73.5	74.6	76.9	(33)
		MCDB	86.7	86.1	85.1	82.1	75.2	74.3	74.7	78.9	80.6	83.8	83.7	85.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	19.2	19.7	19.0	17.4	17.7	19.7	22.1	21.9	20.3	20.5	19.4	(35)
		MCDBR	22.8	23.5	23.4	22.2	21.3	21.1	23.9	26.5	28.2	26.1	24.9	22.9	(36)
	5% WB	MCWBR	8.3	8.5	8.3	8.6	10.1	9.7	10.9	10.6	10.2	9.9	9.7	9.0	(37)
		MCWBR	19.1	19.5	19.5	19.1	17.6	18.5	21.1	23.6	23.5	22.5	20.6	19.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.402	0.397	0.389	0.384	0.358	0.388	0.387	0.469	0.566	0.475	0.417	0.414	(40)
		taud	2.425	2.438	2.453	2.437	2.479	2.391	2.371	2.096	1.851	2.171	2.339	2.369	(41)
	Ebn at Noon		299	295	289	275	269	251	256	244	234	270	292	296	(42)
		Edh at Noon	40	38	36	33	29	30	32	46	64	49	43	42	(43)
(44) All-Sky Solar Radiation	RadAvg		2049	1878	1712	1410	1044	904	1022	1256	1436	1648	2000	2061	(44)
	RadStd		162	149	115	130	147	96	121	92	155	188	189	187	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	+1.96	+2.41	N/A	N/A	N/A	N/A	
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page